



University of Copenhagen
MSc. Physics

Continuous Time Finance 2

Hand-In 1

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1 – Moments

1.a

We aim to solve the SDE:

$$dX = \kappa(\theta - X(t))dt + \sigma dW(t)$$

Consider $Z(t) = e^{\kappa t}X(t)$. Then we have:

$$\begin{aligned} dZ &= \frac{\partial Z}{\partial t}dt + \frac{\partial Z}{\partial X}dX + \frac{\partial^2 Z}{\partial X^2}dX^2 \\ &= \kappa Zdt + e^{\kappa t}(\kappa(\theta - X(t))dt + \sigma dW(t)) \\ &= \kappa e^{\kappa t}X(t)dt + \kappa\theta e^{\kappa t}dt - \kappa e^{\kappa t}X(t)dt + \sigma e^{\kappa t}dW(t) \\ &= \kappa\theta e^{\kappa t}dt + \sigma e^{\kappa t}dW(t) \end{aligned}$$

Integrating both sides from t to $t+u$:

$$\begin{aligned} \int_t^{t+u} dZ &= \kappa\theta \int_t^{t+u} e^{\kappa s}ds + \sigma \int_t^{t+u} e^{\kappa s}dW(s) \\ Z(t+u) - Z(t) &= \theta(e^{\kappa(t+u)} - e^{\kappa t}) + \sigma \int_t^{t+u} e^{\kappa s}dW(s) \\ Z(t+u) &= Z(t) + \theta(e^{\kappa(t+u)} - e^{\kappa t}) + \sigma \int_t^{t+u} e^{\kappa s}dW(s) \end{aligned}$$

Now substitute $Z(t) = e^{\kappa t}X(t)$:

$$\begin{aligned} e^{\kappa(t+u)}X(t+u) &= e^{\kappa t}X(t) + \theta(e^{\kappa(t+u)} - e^{\kappa t}) + \sigma \int_t^{t+u} e^{\kappa s}dW(s) \\ X(t+u) &= e^{-\kappa u}X(t) + \theta(1 - e^{-\kappa u}) + e^{-\kappa(t+u)}\sigma \int_t^{t+u} e^{\kappa s}dW(s) \end{aligned}$$

Taking the expectation over the filtration $\mathcal{F}_t$ yields:

$$\mathbb{E}[X(t+u)|\mathcal{F}_t] = X(t)e^{-\kappa u} + \theta(1 - e^{-\kappa u})$$

We use Björk's Lemma 4.18 to find:

$$\begin{aligned} \text{Var}[X(t+u)|\mathcal{F}_t] &= e^{-2\kappa(t+u)}\sigma^2 \int_t^{t+u} e^{2\kappa s}ds \\ &= \frac{\sigma^2}{2\kappa}e^{-2\kappa(t+u)}(e^{2\kappa(t+u)} - e^{2\kappa t}) \\ &= \frac{\sigma^2}{2\kappa}(1 - e^{-2\kappa u}) \end{aligned}$$

Thus we may conclude:

$$X(t+u)|\mathcal{F}_t \sim \mathcal{N}\left[X(t)e^{-\kappa u} + \theta(1 - e^{-\kappa u}), \frac{\sigma^2(1 - e^{-2\kappa u})}{2\kappa}\right]$$

1.b

We now consider the SDE:

$$dX(t) = \kappa(\theta - X(t))dt + \sigma\sqrt{X(t)}dW(t)$$

To determine $m(t) = \mathbb{E}[X(t)]$, we integrate both sides, take the expectation and interchange:

$$X(t) - X(0) = \int_0^t \kappa(\theta - X(s))ds + \int_0^t \sigma\sqrt{X(s)}dW(s)$$

$$m(t) = X(0) + \int_0^t \kappa(\theta - m(s))ds$$

We may now differentiate, and solve the equation like a regular ODE.

$$m'(t) = \kappa(\theta - m(t))$$

$$m'(t) + \kappa m(t) = \kappa\theta$$

Multiply both sides by $e^{\kappa t}$ and recognize the chain rule.

$$m'(t)e^{\kappa t} + \kappa m(t)e^{\kappa t} = \kappa\theta e^{\kappa t}$$

$$\frac{d}{dt} (m(t)e^{\kappa t}) = \kappa\theta e^{\kappa t}$$

Integrating now, lets us find $m(t)$:

$$m(t)e^{\kappa t} = \int \kappa\theta e^{\kappa s} ds$$

$$m(t) = e^{-\kappa t}\theta e^{\kappa t} + e^{-\kappa t}C$$

$$m(t) = \theta + e^{-\kappa t}C$$

From the initial condition $m(0) = X(0)$ we find:

$$X(0) = \theta + C$$

$$C = X(0) - \theta \rightarrow m(t) = X(0)e^{-\kappa t} + \theta(1 - e^{-\kappa t})$$

1.c

We want to find an ODE for the second moment $h(t) = \mathbb{E}[X^2(t)]$, with the same SDE as in the previous exercise. To achieve this, we consider the process $Z(t) = X^2(t)$:

$$\begin{aligned} dZ &= 2XdX + dX^2 \\ &= 2\kappa X(t)(\theta - X(t))dt + 2(X(t))^{\frac{3}{2}}dW(t) + \sigma^2 X(t)dt \\ &= X(t)(2\kappa\theta - 2\kappa X(t) + \sigma^2)dt + 2(X(t))^{\frac{3}{2}}dW(t) \end{aligned}$$

We now integrate, take the expectation and interchange.

$$\begin{aligned} h(t) &= Z(0) + \int_0^t (2\kappa\theta m(s) - 2\kappa h(s) + \sigma^2 m(s))ds \\ h'(t) &= 2\kappa\theta m(t) - 2\kappa h(t) + \sigma^2 m(t) \end{aligned}$$

Thus we have found the desired ODE for $h(t)$. To determine the variance of $X(t)$, we must first solve this ODE. We proceed as we did in the previous exercise:

$$\begin{aligned} h'(t) + 2\kappa h(t) &= (2\kappa\theta + \sigma^2)m(t) \\ e^{2\kappa t}h'(t) + e^{2\kappa t}2\kappa h(t) &= e^{2\kappa t}(2\kappa\theta + \sigma^2)m(t) \\ \frac{d}{dt}(e^{2\kappa t}h(t)) &= e^{2\kappa t}(2\kappa\theta + \sigma^2)m(t) \\ e^{2\kappa t}h(t) &= \int e^{2\kappa s}(2\kappa\theta + \sigma^2)m(s)ds \end{aligned}$$

Substituting $m(t) = X(0)e^{-\kappa t} + \theta(1 - e^{-\kappa t})$, lets us solve the above integral:

$$\begin{aligned} h(t) &= e^{-2\kappa t}(2\kappa\theta + \sigma^2) \int e^{2\kappa s}(X(0)e^{-\kappa s} + \theta(1 - e^{-\kappa s}))ds \\ &= e^{-2\kappa t}(2\kappa\theta + \sigma^2) \int (X(0)e^{\kappa s} + \theta(e^{2\kappa s} - e^{\kappa s}))ds \\ &= e^{-2\kappa t}(2\kappa\theta + \sigma^2) \left(\frac{X(0)}{\kappa}e^{\kappa t} + \frac{\theta}{2\kappa}e^{2\kappa t} - \frac{\theta}{\kappa}e^{\kappa t} + C \right) \\ &= (2\kappa\theta + \sigma^2) \left(\frac{X(0)}{\kappa}e^{-\kappa t} + \frac{\theta}{2\kappa} - \frac{\theta}{\kappa}e^{-\kappa t} + Ce^{-2\kappa t} \right) \end{aligned}$$

Use initial condition $h(0) = X^2(0)$:

$$\begin{aligned} X^2(0) &= (2\kappa\theta + \sigma^2) \left(\frac{X(0)}{\kappa} + \frac{\theta}{2\kappa} - \frac{\theta}{\kappa} + C \right) \\ C &= \frac{X^2(0)}{2\kappa\theta + \sigma^2} - \frac{X(0)}{\kappa} - \frac{\theta}{2\kappa} + \frac{\theta}{\kappa} \end{aligned}$$

We may now finally fully determine $h(t)$:

$$\begin{aligned} h(t) &= (2\kappa\theta + \sigma^2) \left(\frac{X(0)}{\kappa}e^{-\kappa t} + \frac{\theta}{2\kappa} - \frac{\theta}{\kappa}e^{-\kappa t} + \left(\frac{X^2(0)}{2\kappa\theta + \sigma^2} - \frac{X(0)}{\kappa} - \frac{\theta}{2\kappa} + \frac{\theta}{\kappa} \right) e^{-2\kappa t} \right) \\ &= X^2(0)e^{-2\kappa t} + (2\kappa\theta + \sigma^2) \left(\frac{X(0)}{\kappa}(e^{-\kappa t} - e^{-2\kappa t}) + \frac{\theta}{2\kappa}(1 - e^{-2\kappa t}) + \frac{\theta}{\kappa}(e^{-2\kappa t} - e^{-\kappa t}) \right) \\ &= X^2(0)e^{-2\kappa t} + (2\kappa\theta + \sigma^2) \left(\frac{X(0)}{\kappa}(e^{-\kappa t} - e^{-2\kappa t}) + \frac{\theta}{2\kappa}(1 + e^{-2\kappa t} - 2e^{-\kappa t}) \right) \\ &= X^2(0)e^{-2\kappa t} + 2X(0)\theta(e^{-\kappa t} - e^{-2\kappa t}) + \frac{X(0)\sigma^2}{\kappa}\theta(e^{-\kappa t} - e^{-2\kappa t}) + \theta^2(1 + e^{-2\kappa t} - 2e^{-\kappa t}) \\ &\quad + \frac{\theta\sigma^2}{2\kappa}(1 + e^{-2\kappa t} - 2e^{-\kappa t}) \end{aligned}$$

Computing the variance of $X(t)$:

$$\begin{aligned}\text{Var}[X(t)] &= \mathbb{E}[X^2(t)] - \mathbb{E}[X(t)]^2 \\ &= h(t) - m^2(t)\end{aligned}$$

It is pretty straightforward to find $m^2(t)$:

$$m^2(t) = X^2(0)e^{-2\kappa t} + \theta^2(1 + e^{-2\kappa t} - 2e^{-\kappa t}) + 2X(0)\theta(e^{-\kappa t} - e^{-2\kappa t})$$

Thus we finally arrive at an expression for the variance:

$$\begin{aligned}\text{Var}[X(t)] &= X^2(0)e^{-2\kappa t} + 2X(0)\theta(e^{-\kappa t} - e^{-2\kappa t}) + \frac{X(0)\sigma^2}{\kappa}\theta(e^{-\kappa t} - e^{-2\kappa t}) + \theta^2(1 + e^{-2\kappa t} - 2e^{-\kappa t}) \\ &\quad + \frac{\theta\sigma^2}{2\kappa}(1 + e^{-2\kappa t} - 2e^{-\kappa t}) - X^2(0)e^{-2\kappa t} - \theta^2(1 + e^{-2\kappa t} - 2e^{-\kappa t}) - 2X(0)\theta(e^{-\kappa t} - e^{-2\kappa t}) \\ &= \frac{X(0)\sigma^2}{\kappa}\theta(e^{-\kappa t} - e^{-2\kappa t}) + \frac{\theta\sigma^2}{2\kappa}(1 + e^{-2\kappa t} - 2e^{-\kappa t}) \\ &= \frac{X(0)\sigma^2}{\kappa}\theta(e^{-\kappa t} - e^{-2\kappa t}) + \frac{\theta\sigma^2}{2\kappa}(1 - e^{-\kappa t})^2\end{aligned}$$

2 – The Bachelier Model

2.a

We consider the Bachelier model with an interest rate $= 0$. We aim to find the arbitrage-free price for a european call-option, i.e. the price of a call-option, with the underlying stock being of the form $dS(t) = \sigma dW(t)$, which means according to Björk's Lemma 4.18:

$$\mathbb{E}[S(T)] = S(t)$$

$$\text{Var}[S(T)] = \sigma^2(T - t)$$

Now define the process $Z(T) = S(T) - K$. It is trivial to see that $Z \sim \mathcal{N}[S(t) - K, \sigma^2(T - t)]$.

Using the hint, that was generously provided in the assignment text:

$$\mathbb{E}[Z \cdot \mathbb{1}_{0 \leq Z \leq \infty}] = (S(t) - K) \left(\Phi(\infty) - \Phi\left(\frac{K - S(t)}{\sigma\sqrt{T-t}}\right) \right) + \sigma\sqrt{T-t} \left(\phi\left(\frac{K - S(t)}{\sigma\sqrt{T-t}}\right) - \phi(\infty) \right)$$

Now rewriting the CDF part by rearranging the limits when subtracting, considering symmetries of the normal distribution $\phi(x) = \phi(-x)$, and applying $\phi(\infty) = 0$:

$$\begin{aligned} \mathbb{E}[Z \cdot \mathbb{1}_{0 \leq Z \leq \infty}] &= (S(t) - K) \int_{\frac{K - S(t)}{\sigma\sqrt{T-t}}}^{\infty} \phi(x) dx + \sigma\sqrt{T-t} \phi\left(\frac{K - S(t)}{\sigma\sqrt{T-t}}\right) \\ &= (S(t) - K) \int_{-\infty}^{\frac{S(t) - K}{\sigma\sqrt{T-t}}} \phi(x) dx + \sigma\sqrt{T-t} \phi\left(\frac{S(t) - K}{\sigma\sqrt{T-t}}\right) \\ &= (S(t) - K) \Phi\left(\frac{S(t) - K}{\sigma\sqrt{T-t}}\right) + \sigma\sqrt{T-t} \phi\left(\frac{S(t) - K}{\sigma\sqrt{T-t}}\right) \end{aligned}$$

Now using Björk's proposition 5.5 (Feynman-Kac):

$$\Pi_{\text{call}}^{\text{bach}}(t) = \mathbb{E}[Z \cdot \mathbb{1}_{0 \leq Z \leq \infty}] = (S(t) - K) \Phi\left(\frac{S(t) - K}{\sigma\sqrt{T-t}}\right) + \sigma\sqrt{T-t} \phi\left(\frac{S(t) - K}{\sigma\sqrt{T-t}}\right)$$

To determine Δ we simply differentiate and apply the chain rule:

$$\begin{aligned} \Delta_{\text{call}}^{\text{bach}}(t) &= \frac{\partial \Pi_{\text{call}}^{\text{bach}}}{\partial s} = \frac{\partial}{\partial s} \left(S(t) \Phi\left(\frac{S(t) - K}{\sigma\sqrt{T-t}}\right) \right) - \frac{\partial}{\partial s} \left(K \Phi\left(\frac{S(t) - K}{\sigma\sqrt{T-t}}\right) \right) + \frac{\partial}{\partial s} \left(\sigma\sqrt{T-t} \phi\left(\frac{S(t) - K}{\sigma\sqrt{T-t}}\right) \right) \\ &= \Phi\left(\frac{S(t) - K}{\sigma\sqrt{T-t}}\right) + \frac{S(t)}{\sigma\sqrt{T-t}} \phi\left(\frac{S(t) - K}{\sigma\sqrt{T-t}}\right) - \frac{K}{\sigma\sqrt{T-t}} \phi\left(\frac{S(t) - K}{\sigma\sqrt{T-t}}\right) \\ &\quad - \frac{S(t) - K}{\sigma\sqrt{T-t}} \phi\left(\frac{S(t) - K}{\sigma\sqrt{T-t}}\right) \\ &= \Phi\left(\frac{S(t) - K}{\sigma\sqrt{T-t}}\right) \end{aligned}$$

We are now tasked to find the arbitrage-free call-option price for a non-zero interest rate, i.e. for the underlying process:

$$dS = rS(t)dt + \sigma dW(t)$$

We solve this problem just as in 1.a. Let $Z(t) = e^{-rt}S(t)$:

$$dZ = -rZ(t)dt + re^{-rt}S(t)dt + \sigma e^{-rt}dW(t)$$

$$dZ = -re^{-rt}S(t)dt + re^{-rt}S(t)dt + \sigma e^{-rt}dW(t)$$

$$Z(T) = Z(t) + \sigma \int_t^T e^{-ru}dW(u)$$

$$S(T) = S(t)e^{r(T-t)} + \sigma e^{rT} \int_t^T e^{-ru}dW(u)$$

Finding the mean is straightforward and we can determine the variance by once again invoking Björk's Lemma 4.18:

$$\mathbb{E}[S(T)] = S(t)e^{r(T-t)}$$

$$\begin{aligned} \text{Var}[S(T)] &= \sigma^2 e^{2rT} \int_t^T e^{-2ru} du \\ &= \frac{\sigma^2 e^{2rT}}{2r} (e^{-2rt} - e^{-2rT}) \\ &= \frac{\sigma^2}{2r} (e^{2r(T-t)} - 1) \end{aligned}$$

Thus $S(T) \sim \mathcal{N}[S(t)e^{r(T-t)}, \frac{\sigma^2}{2r}(e^{2r(T-t)} - 1)]$ and we use the hint provided once again. Let $Y(T) = S(T) - K$ so that $Y \sim \mathcal{N}[S(t)e^{r(T-t)} - K, \frac{\sigma^2}{2r}(e^{2r(T-t)} - 1)]$.

$$\begin{aligned} \mathbb{E}[Y \cdot \mathbb{1}_{0 \leq Y \leq \infty}] &= (S(t)e^{r(T-t)} - K) \left(\Phi(\infty) - \Phi\left(\frac{\sqrt{2r}(K - S(t)e^{r(T-t)})}{\sigma\sqrt{e^{2r(T-t)} - 1}}\right) \right) \\ &\quad + \frac{\sigma\sqrt{e^{2r(T-t)} - 1}}{\sqrt{2r}} \left(\phi\left(\frac{\sqrt{2r}(K - S(t)e^{r(T-t)})}{\sigma\sqrt{e^{2r(T-t)} - 1}}\right) - \phi(\infty) \right) \\ &= (S(t)e^{r(T-t)} - K) \Phi\left(\frac{\sqrt{2r}(S(t)e^{r(T-t)} - K)}{\sigma\sqrt{e^{2r(T-t)} - 1}}\right) \\ &\quad + \frac{\sigma\sqrt{e^{2r(T-t)} - 1}}{\sqrt{2r}} \phi\left(\frac{\sqrt{2r}(S(t)e^{r(T-t)} - K)}{\sigma\sqrt{e^{2r(T-t)} - 1}}\right) \end{aligned}$$

Where we used the same techniques as for the calculations for interest-rate-free asset, to simplify the expression. Taking the discounted expectation then gives us the price of the call-option (Björk's Prop. 5.6):

$$\begin{aligned} \Pi_{\text{call}}^{\text{bach}}(t) &= e^{-r(T-t)} \mathbb{E}[Z \cdot \mathbb{1}_{0 \leq Z \leq \infty}] = e^{-r(T-t)} \left[(S(t)e^{r(T-t)} - K) \Phi\left(\frac{\sqrt{2r}(S(t)e^{r(T-t)} - K)}{\sigma\sqrt{e^{2r(T-t)} - 1}}\right) \right. \\ &\quad \left. + \frac{\sigma\sqrt{e^{2r(T-t)} - 1}}{\sqrt{2r}} \phi\left(\frac{\sqrt{2r}(S(t)e^{r(T-t)} - K)}{\sigma\sqrt{e^{2r(T-t)} - 1}}\right) \right] \end{aligned}$$

2.b

The python code used in these exercises is given in the appendix as well as submitted to absalon. For the given set of parameters, we started by computing the call-prices.

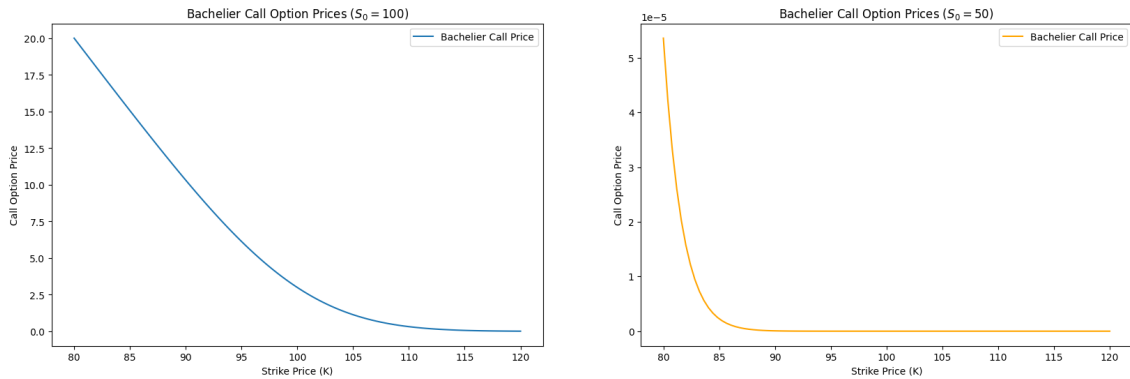


Figure 1: Call-prices computed using the Bachelier model. Note the very small values on the y-axis for $S_0 = 50$, which is likely caused by the given strike prices being significantly out of the money for the stock.

We then used a simple bisection algorithm to determine the implied volatility, that would match the Black-Scholes price to the computed Bachelier price above. Below are the resulting graphs:

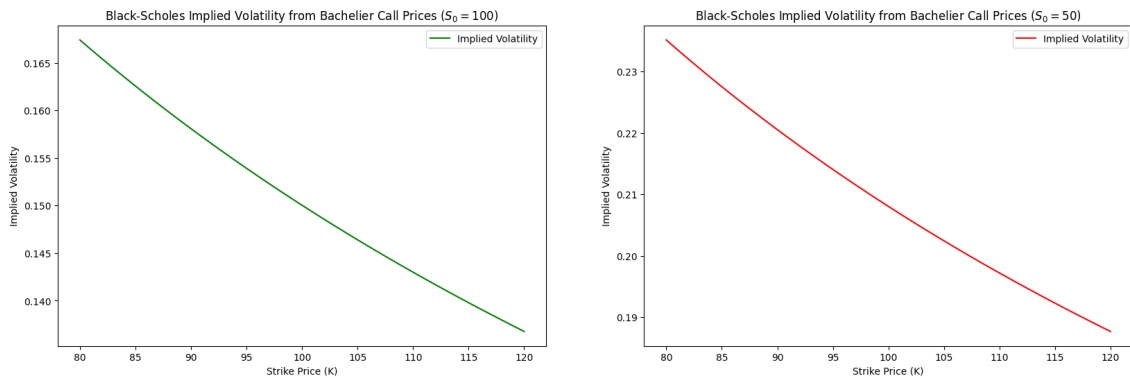


Figure 2: Black-Scholes implied volatility for Bachelier Call prices. Obtained using a bisection algorithm.

The implied volatilities are evidently not uniform and have a rather negative trend with increasing strike prices. They are also significantly smaller than the $\sigma = 15$, used to calculate the call-prices. The latter observation is likely due to how the two models treat volatility: In the Bachelier model volatility is an absolute amount of stock value, while in the Black-Scholes model it represents a fraction of the stock value. Thus we also see the implied volatility increasing for $S_0 = 50$, because $\sigma = 15$ represents a much larger fractional fluctuation of the total stock value, than for the $S_0 = 100$ case. Furthermore, The downward trend of the graphs suggests, that the Bachelier model assigns higher probabilities to positive payoffs for low-strike prices, than the Black-Scholes model does. This makes sense, since the constant volatility of the Bachelier model allows stocks that have low prices to fluctuate just as much as higher valued stocks, resulting in a higher probability that the call-option, despite having a rough start, ends up in the money.

3– Quanto Hedging and The Kingdom of Denmark Put

Consider a US bank account B_{US} , a Japanese bank account B_J , a Japanese stock S_J traded in Yen and an exchange rate X , all governed by the following dynamics under the US-Martingale measure:

$$\begin{aligned} dB_{US}(t) &= r_{US}B_{US}(t)dt \\ dX(t) &= X(t)(r_{US} - r_J)dt + X(t)\sigma_X^\top dW(t) \\ dB_J(t) &= r_JB_J(t)dt \\ dS_J(t) &= S_J(t)(r_J - \sigma_X^\top \sigma_J)dt + S_J(t)\sigma_J^\top dW(t) \end{aligned}$$

Furthermore, consider a Quanto put of the form:

$$\mathcal{X}_{\text{quanto}} = Y_0(K - S_J(T))^+$$

Where Y_0 is some previously agreed upon exchange rate.

3.a

We aim to find the pricing function for the quanto put $F(S_J(t), t)$. A simple way to do this is to recognize the SDE for S_J looks the same as the SDE for a stock with a constant dividend yield:

$$dS(t) = (r - \delta)S(t)dt + \sigma S(t)dW(t)$$

We know that the call-price looks like for BS-model(from lectures):

$$\begin{aligned} \text{call} &= e^{-\delta(T-t)}S\Phi(d_1) - e^{-r(T-t)}K\Phi(d_2) \\ d_{1,2} &= \frac{\ln(\frac{S}{K}) + ((r - \delta) \pm ((r - \delta) \pm \frac{1}{2}\sigma^2)(T - t))}{\sigma\sqrt{T - t}} \end{aligned}$$

Now use the put-call parity:

$$\begin{aligned} \text{put} &= -e^{-\delta(T-t)}S + e^{-r(T-t)}K + \text{call} \\ \text{put} &= -e^{-\delta(T-t)}S + e^{-r(T-t)}K + e^{-\delta(T-t)}S\Phi(d_1) - e^{-r(T-t)}K\Phi(d_2) \\ \text{put} &= -e^{-\delta(T-t)}S + e^{-r(T-t)}K + e^{-\delta(T-t)}S\Phi(d_1) - e^{-r(T-t)}K\Phi(d_2) \\ \text{put} &= -e^{-\delta(T-t)}S\Phi(-d_1) + e^{-r(T-t)}K\Phi(-d_2) \end{aligned}$$

Where we used $1 - \Phi(x) = \Phi(-x)$ Now multiply by Y_0 to replicate the quanto-put and substitute $r = r_{US}$ and $\delta = r_{US} - r_J + \sigma_X^\top \sigma_J$.

$$\begin{aligned} F^{QP} &= Y_0(e^{-r_{US}-r_J+\sigma_X^\top\sigma_J(T-t)}S\Phi(-d_1) - e^{-r_{US}(T-t)}K\Phi(-d_2)) \\ F^{QP} &= Y_0e^{-r_{US}(T-t)}(e^{r_J-\sigma_X^\top\sigma_J(T-t)}S\Phi(-d_1) - K\Phi(-d_2)) \end{aligned}$$

Thus we have found the arbitrage-free pricing function of the quanto put. Differentiating is straightforward:

$$\begin{aligned} \frac{\partial F^{QP}}{\partial s} &= Y_0e^{-r_{US}(T-t)} \left[-K \frac{\partial d_2}{\partial s} \phi(-d_2) - e^{(r_J-\sigma_X^\top\sigma_J)(T-t)} \left(\Phi(-d_1) - s \frac{\partial d_1}{\partial s} \phi(-d_1) \right) \right] \\ &= Y_0e^{-r_{US}(T-t)} \left[-\frac{K}{s\sqrt{T-t}||\sigma_J||} \phi(-d_2) - e^{(r_J-\sigma_X^\top\sigma_J)(T-t)} \left(\Phi(-d_1) - \frac{1}{\sqrt{T-t}||\sigma_J||} \phi(-d_1) \right) \right] \end{aligned}$$

I'm not sure how to proceed from this point on.

3.b

Drawing inspiration from the template code provided [2], we implemented a hedge simulator in python. Some minor changes were made for improved runtimes, such as calculating stochastic trajectories all at once (and then iterating over a list instead) and computing all dot products before running the simulation. We first did an example run with $n = 252$ rebalancing steps, and plotted the ideal payoff on top. As seen in the below figure, the points are generally spread far from the ideal payoff, which strongly suggests that this hedging strategy does not replicate the payoff of the quanto-put well. This pattern persists for a higher frequency of rebalancing steps, as seen on the righthand-side plot, where the MSE flatlines at about ~ 440 . Through induction we may reasonable conclude that this means, that the hedging error cannot be offset with more rebalancing steps (even in the $n \rightarrow \infty$ limit), and is caused by an intrinsic fault in the trading strategy.

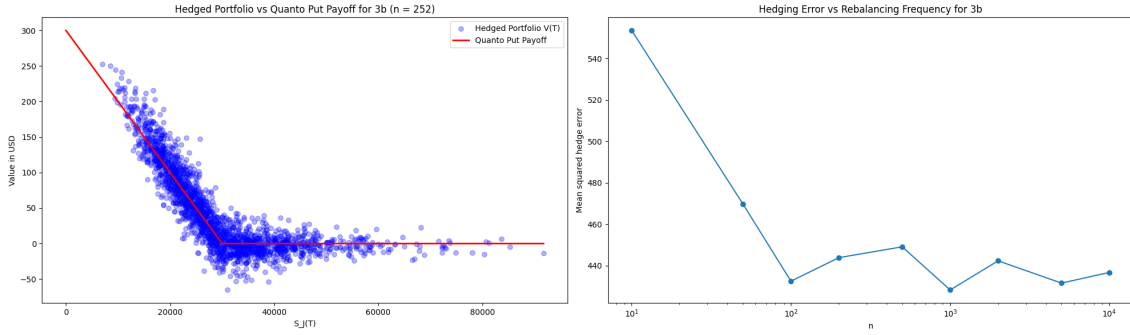


Figure 3

3.c

In contrast to the previous trading strategy, the points now fit much better on the put payoff graph, for $n = 252$. The righthand-side plot also shows a clear systematic decrease of the MSE with an increasing amount of rebalancing steps. Once again we may inductively conclude that this portfolio replicates the quanto put properly, and that the MSE tends to 0 as $n \rightarrow \infty$.

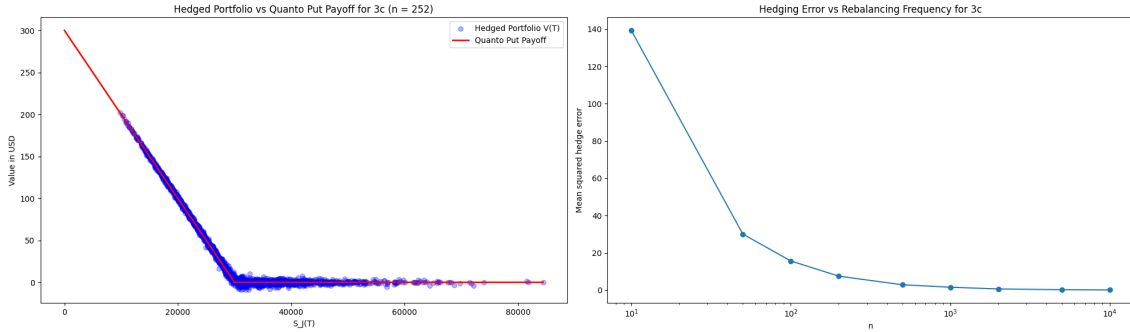


Figure 4

3.d

Consider the pricing function PDE:

$$F_t + (r_J - \sigma_X \sigma_J^\top) S_J F_s + \frac{1}{2} |\sigma_J|^2 S_J^2 F_{ss} = r_{US} F$$

substitute that into the SDE:

$$\begin{aligned} dF &= F_t dt + F_s dS_J + \frac{1}{2} F_{ss} dS_J^2 \\ &= F_t dt + F_s S_J (r_J - \sigma_X^\top \sigma_J) dt + F_s S_J \sigma_J^\top dW + \frac{1}{2} F_{ss} S_J^2 |\sigma_J|^2 dt \\ &= r_{US} F + F_s S_J \sigma_J^\top dW \end{aligned}$$

In order for a portfolio to replicate the put, its value function must fulfill the above PDE. Now consider the portfolio from 3b, which consists of a foreign stock position and domestic bank account:

$$\begin{aligned} V &= h_s S_J X + B_{US} \\ dV &= h_s d(S_J X) + r_{US} B_{US} dt \\ &= h_s (S_J dX + dS_J X + dS_J dX) + r_{US} B_{US} dt \\ &= h_s (S_J X (r_{US} - r_J) dt + S_J X \sigma_X^\top dW + X S_J (r_J - \sigma_X^\top \sigma_J) dt + X S_J \sigma_J^\top dW + S_J X \sigma_X^\top \sigma_J dt) \\ &\quad + r_{US} B_{US} dt \\ &= h_s (S_J X r_{US} dt + S_J X (\sigma_X^\top + \sigma_J^\top) dW) + r_{US} B_{US} dt \\ &= \frac{F_s}{X} (S_J X r_{US} dt + S_J X (\sigma_X^\top + \sigma_J^\top) dW) + r_{US} B_{US} dt \\ &= F_s S_J r_{US} dt + S_J F_s (\sigma_X^\top + \sigma_J^\top) dW + r_{US} B_{US} dt \\ &= r_{US} (F_s S_J + B_{US}) dt + S_J F_s (\sigma_X^\top + \sigma_J^\top) dW \\ &= r_{US} V dt + S_J F_s (\sigma_X^\top + \sigma_J^\top) dW \end{aligned}$$

We can now see clearly that the portfolio from 3.b does not replicate (In particular because of the diffusion term containing $\sigma_X^\top$). The portfolio in 3.c also includes a foreign bank account, allowing for proper replication:

$$\begin{aligned} V &= h_s S_J X + B_{US} + h_X X B_J \\ dV &= h_s d(S_J X) + r_{US} B_{US} dt + h_X d(X B_J) \\ &= r_{US} (F_s S_J + B_{US}) dt + S_J F_s (\sigma_X^\top + \sigma_J^\top) dW + h_X B_J dX + h_X X dB_J \\ &= r_{US} (F_s S_J + B_{US}) dt + S_J F_s (\sigma_X^\top + \sigma_J^\top) dW + h_X B_J (X (r_{US} - r_J) dt + X \sigma_X^\top dW) + h_X r_J X B_J dt \\ &= r_{US} (F_s S_J + B_{US}) dt + S_J F_s (\sigma_X^\top + \sigma_J^\top) dW + h_X B_J (X r_{US} dt + X \sigma_X^\top dW) \\ &= r_{US} (F_s S_J + B_{US}) dt + S_J F_s (\sigma_X^\top + \sigma_J^\top) dW + \left(-\frac{F_s S_J}{X B_J} \right) B_J (X r_{US} dt + X \sigma_X^\top dW) \\ &= r_{US} B_{US} dt + S_J F_s \sigma_J^\top dW \\ &= r_{US} V dt + S_J F_s \sigma_J^\top dW \end{aligned}$$

Note that the last equality follows from the fact that $h_s S_J X + h_X X B_J = 0$, when the hedge is properly balanced. Thus we see that the portfolio from 3.c obeys the pricing PDE and is properly replicating, unlike the portfolio from 3.b.

3.e

Unfortunately I did not manage my time well enough to complete this exercise.

References

- [1] T. Björk, Arbitrage Theory in Continuous time, 4th edition. Oxford University Press (2020).
- [2] R. Poulsen, Exercise3_QuantoHedge. Dropbox (2026).
URL https://www.dropbox.com/scl/fi/lp182s4d8j02hhilu8dmj/Exercise3_QuantoHedge.R?rlkey=5ps3rvygdejrgxm08udu1bkff&e=1&dl=0

Hand-in 1

February 23, 2026

1 Hand-In 1

```
[5]: import numpy as np
import matplotlib.pyplot as plt
import scipy.stats as stats
```

1.1 2b)

```
[6]: def Bachelier_calls(K, S0, T, sigma):
    d = (S0 - K) / (sigma * np.sqrt(T))
    call_price = (S0 - K) * stats.norm.cdf(d) + sigma * np.sqrt(T) * stats.
    ↪norm.pdf(d)
    return call_price

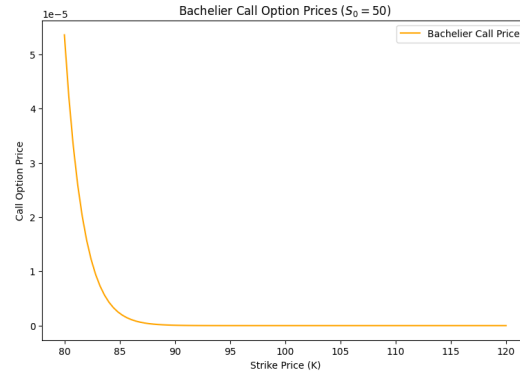
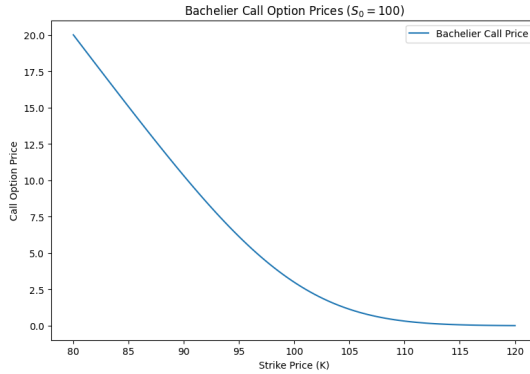
K_vals = np.linspace(80, 120, 100)

call_prices_1 = Bachelier_calls(K_vals, S0 = 100, T = 0.25, sigma = 15)
call_prices_2 = Bachelier_calls(K_vals, S0 = 50, T = 0.25, sigma = 15)

fig, ax = plt.subplots(1, 2, figsize=(20, 6))
ax[0].plot(K_vals, call_prices_1, label = 'Bachelier Call Price')
ax[0].set_title('Bachelier Call Option Prices ($S_0=100$)')
ax[0].set_xlabel('Strike Price (K)')
ax[0].set_ylabel('Call Option Price')
ax[0].legend()

ax[1].plot(K_vals, call_prices_2, label = 'Bachelier Call Price',
    ↪color='orange')
ax[1].set_title('Bachelier Call Option Prices ($S_0=50$)')
ax[1].set_xlabel('Strike Price (K)')
ax[1].set_ylabel('Call Option Price')
ax[1].legend()

plt.show()
```



```
[7]: def BlackScholes_calls(K, S0, T, sigma):
    d1 = (np.log(S0 / K) + (sigma**2 / 2) * T) / (sigma * np.sqrt(T))
    d2 = d1 - sigma * np.sqrt(T)
    call_price = S0 * stats.norm.cdf(d1) - K * stats.norm.cdf(d2)
    return call_price

#Bisection method to find implied volatility
def implied_volatility_bisection(K, S0, T, market_price, tol=1e-30,
    ↪max_iter=1000):
    low_vol = 0.1
    high_vol = 5.0

    for i in range(max_iter):
        mid_vol = (low_vol + high_vol) / 2
        price = BlackScholes_calls(K, S0, T, mid_vol)
        if abs(price - market_price) < tol:
            return mid_vol
        elif price < market_price:
            low_vol = mid_vol
        else:
            high_vol = mid_vol
    return mid_vol

implied_vols_1 = [implied_volatility_bisection(K, 100, 0.25, price) for K,
    ↪price in zip(K_vals, call_prices_1)]
implied_vols_2 = [implied_volatility_bisection(K, 50, 0.25, price) for K, price
    ↪in zip(K_vals, call_prices_2)]

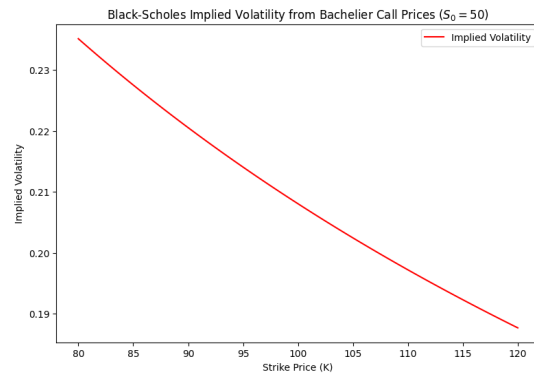
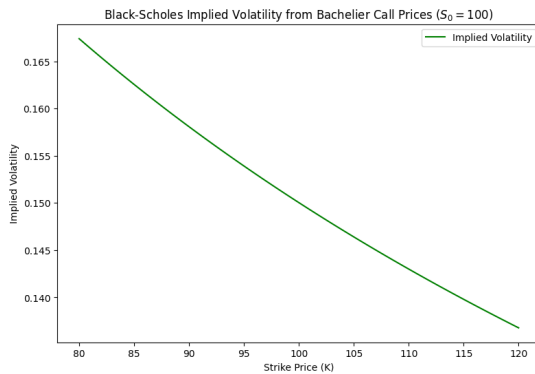
fig, ax = plt.subplots(1, 2, figsize=(20, 6))
ax[0].plot(K_vals, implied_vols_1, label = 'Implied Volatility', color='green')
ax[0].set_title('Black-Scholes Implied Volatility from Bachelier Call Prices
    ↪($S_0=100$)')
ax[0].set_xlabel('Strike Price (K)')
```

```

ax[0].set_ylabel('Implied Volatility')
ax[0].legend()

ax[1].plot(K_vals, implied_vols_2, label = 'Implied Volatility', color='red')
ax[1].set_title('Black-Scholes Implied Volatility from Bachelier Call Prices_
↳($S_0=50$)')
ax[1].set_xlabel('Strike Price (K)')
ax[1].set_ylabel('Implied Volatility')
ax[1].legend()
plt.show()

```



1.2 3b)

```

[13]: #Global parameters
r_US = 0.03
r_J = 0
sig_X = np.array([0.1, 0.02])
sig_J = np.array([0, 0.25])

S_J0 = 30000
X_0 = 0.01
Y_0 = X_0
K = S_J0
T_max = 2

#Calculate constants for faster runtimes
dot_XJ = sig_X @ sig_J
norm_J = np.linalg.norm(sig_J)
dot_XX = sig_X @ sig_X
dot_JJ = sig_J @ sig_J

#Pricing and hedging functions

```

```

def F_QP(t, s, T = T_max):
    d1 = (np.log(s/K) + (r_J - dot_XJ + norm_J**2 / 2) * (T - t))/(norm_J * np.
    ↪sqrt(T - t))
    d2 = d1 - norm_J * np.sqrt(T - t)
    F = Y_0 * np.exp(-r_US * (T - t)) * (K * stats.norm.cdf(-d2) - s * np.
    ↪exp((r_J - dot_XJ) * (T - t)) * stats.norm.cdf(-d1))
    return F

def dF_QPds(t, s, T = T_max):
    d1 = (np.log(s/K) + (r_J - dot_XJ + norm_J**2 / 2) * (T - t))/(norm_J * np.
    ↪sqrt(T - t))
    dFds = Y_0 * np.exp((r_J - dot_XJ - r_US) * (T - t)) * (stats.norm.cdf(d1)
    ↪- 1)
    return dFds

def Hedge_simulator(hedge_type, n):
    m = 2000 #Number of simulations
    dt = T_max / n
    results = np.zeros((m, 3))

    if hedge_type in ['3b', '3c']:
        mu_X = r_US - r_J
        mu_J = r_J - dot_XJ

    for i in range(m):
        #Initialize simulation
        S_J = S_J0
        X = X_0
        Vpf = F_QP(0, S_J)
        hS_J = dF_QPds(0, S_J)/X
        if hedge_type == '3b':
            hX = 0 #Hedging on currency is not allowed

        elif hedge_type == '3c':
            hX = -hS_J * S_J #Hedging on currency is allowed

        hB_US = Vpf - hS_J * S_J * X - hX * X

        dW = np.random.normal(0, np.sqrt(dt), size = (2, n))
        X = X * np.cumprod(np.exp((mu_X - 0.5 * dot_XX) * dt + sig_X @ dW))
        S_J = S_J * np.cumprod(np.exp((mu_J - 0.5 * dot_JJ) * dt + sig_J @ dW))

        for j in range(n):
            Vpf = hS_J * S_J[j] * X[j] + hX * X[j] * np.exp(r_J * dt) + hB_US *
            ↪np.exp(r_US * dt)

        #Rebalance the hedge

```

```

        if j < n-1: #No need to rebalance at the last step
            hS_J = dF_QPds((j + 1) * dt, S_J[j])/X[j]
            if hedge_type == '3b':
                hX = 0
            elif hedge_type == '3c':
                hX = -hS_J * S_J[j]
            hB_US = Vpf - hS_J * S_J[j] * X[j] - hX * X[j]

#Save results
        results[i, 0] = Vpf
        results[i, 1] = Y_0 * max(K - S_J[-1], 0)
        results[i, 2] = S_J[-1]

    return results

#Plotting results for 3b
results = Hedge_simulator('3b', 252)
x_axis = np.arange(1, np.floor(max(results[:, 2])) + 1)

fig, ax = plt.subplots(1, 2, figsize=(20, 6))
ax[0].scatter(results[:,2], results[:,0], color='blue', alpha=0.3,
    ↪label='Hedged Portfolio V(T)')
ax[0].plot(x_axis, Y_0*np.maximum(K - x_axis,0), color='red', lw=2,
    ↪label='Quanto Put Payoff')
ax[0].set_xlabel('S_J(T)')
ax[0].set_ylabel('Value in USD')
ax[0].set_title('Hedged Portfolio vs Quanto Put Payoff for 3b (n = %s)' %252)
ax[0].legend()

n_vals = [10, 50, 100, 200, 500, 1000, 2000, 5000, 10000]
errors = []

for n in n_vals:
    results = Hedge_simulator('3b', n)
    hedge_error = results[:,0] - results[:,1]
    errors.append(np.mean(hedge_error**2))

ax[1].plot(n_vals, errors, marker='o')
ax[1].set_xscale('log')
ax[1].set_xlabel('n')
ax[1].set_ylabel('Mean squared hedge error')
ax[1].set_title('Hedging Error vs Rebalancing Frequency for 3b')

plt.tight_layout()
plt.show()

```

```

#Plotting results for 3c
results = Hedge_simulator('3c', 252)
x_axis = np.arange(1, np.floor(max(results[:, 2])) + 1)

fig, ax = plt.subplots(1, 2, figsize=(20, 6))
ax[0].scatter(results[:,2], results[:,0], color='blue', alpha=0.3,
    ↳label='Hedged Portfolio V(T)')
ax[0].plot(x_axis, Y_0*np.maximum(K - x_axis,0), color='red', lw=2,
    ↳label='Quanto Put Payoff')
ax[0].set_xlabel('S_J(T)')
ax[0].set_ylabel('Value in USD')
ax[0].set_title('Hedged Portfolio vs Quanto Put Payoff for 3c (n = %s)' %252)
ax[0].legend()

n_vals = [10, 50, 100, 200, 500, 1000, 2000, 5000, 10000]
errors = []

for n in n_vals:
    results = Hedge_simulator('3c', n)
    hedge_error = results[:,0] - results[:,1]
    errors.append(np.mean(hedge_error**2))

ax[1].plot(n_vals, errors, marker='o')
ax[1].set_xscale('log')
ax[1].set_xlabel('n')
ax[1].set_ylabel('Mean squared hedge error')
ax[1].set_title('Hedging Error vs Rebalancing Frequency for 3c')

plt.tight_layout()
plt.show()

```

